

WELL COMPLETION REPORT

Synthetic Field Cluster — SYN-Assam-Duliajan (Upper-Assam-type setting)

1. WELL IDENTIFICATION

Well ID	W008
Well Name	SYN-Assam-Duliajan-008
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude	27.307921 N
Longitude	95.286728 E
Well Status	Drilled
Well Role	Historical (offset reference well)
Spud Date	24-Mar-2018
Drilling Completed	05-Apr-2018
Report / Completion Date	13-Apr-2018
Total Depth (Driller)	3,690 m
Primary Formation	Sandstone
Formations Penetrated	Claystone / Shale / Sandstone / Limestone
Rig	SYN-RIG-E1400-11
Drilling Days	13 days
Recorded Events	8
Recorded NPT	148.1 hours

2. MACHINE-READABLE HEADER BLOCK

Document Type: Well Completion Report
Well ID: W008
Well Name: SYN-Assam-Duliajan-008
Field: SYN-Assam-Duliajan
Latitude: 27.307921
Longitude: 95.286728
Total Depth: 3,690 m
Primary Formation: Sandstone
Well Status: Drilled
Spud Date: 2018-03-24
Event Count: 8
Total NPT: 148.1 hours

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). The well, coordinates, formations, events and measurements described here are generated for software demonstration and are not an operational record of any company. Location names are used for interface readability only.

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3. WELL SUMMARY

3.1 Objective

Well W008 (SYN-Assam-Duliajan-008) was drilled to provide an additional development penetration of the primary sandstone objective and confirm the area formation sequence. The well was drilled vertically from surface to a driller's total depth of 3,690 m within the SYN-Assam-Duliajan synthetic cluster.

3.2 Drilling Interval

Surface to 3,690 m, completed in 13 rotary drilling days between 24-Mar-2018 and 05-Apr-2018. Average overall penetration rate for the well was 11.8 m/hr on a 24-hour basis.

3.3 Formations Encountered

The well penetrated 4 formation units: Claystone (0–838 m), Shale (838–2,020 m), Sandstone (2,020–3,271 m), Limestone (3,271–3,690 m). The primary objective formation was Sandstone.

3.4 Major Operational Events

A total of 8 reportable events were recorded, comprising 3 × Wellbore Instability, 2 × Cementing Issue, 2 × Mud Loss, 1 × Casing Issue. Aggregate non-productive time was 148.1 hours. Event detail is given in Section 6.

3.5 Casing Operations

4 casing strings were run: 13-3/8" at 840 m, 9-5/8" at 2,020 m, 7" at 3,450 m, 5" liner at 3,690 m. All strings were cemented as detailed in Section 7.

3.6 Completion Status

Well status at report date: **Drilled**. The well was drilled and logged to total depth and suspended pending completion decisions.

4. OFFSET-WELL CONTEXT

The following synthetic wells of the same cluster lie within short offset distance and share a comparable formation sequence. They are the primary comparison set for offset-well analysis.

Well ID	Well Name	Distance (km)	TD (m)	Primary Formation	Events
W003	SYN-Assam-Duliajan-003	4.92	3,850	Sandstone	8
W005	SYN-Assam-Duliajan-005	5.89	3,560	Sandstone	5
W001	SYN-Assam-Duliajan-001	8.34	3,470	Sandstone	4
W002	SYN-Assam-Duliajan-002	9.05	3,450	Sandstone	7
W006	SYN-Assam-Duliajan-006	10.11	3,960	Sandstone	0
W010	SYN-Assam-Duliajan-010	10.75	3,260	Sandstone	7

5. FORMATION / DEPTH RECORD

Formation tops and bases as recorded from cuttings and log correlation. Depths are driller's depths below rotary table.

Unit	Formation	Top Depth (m)	Bottom Depth (m)	Thickness (m)	Remarks
1	Claystone	0	838	838	Wellbore Instability, Cementing Issue recorded (2 events)
2	Shale	838	2,020	1,182	Cementing Issue, Wellbore Instability recorded (2 events)
3	Sandstone	2,020	3,271	1,251	Mud Loss recorded (2 events)
4	Limestone	3,271	3,690	419	Wellbore Instability, Casing Issue recorded (2 events)

Formation: Claystone Top: 0 m Bottom: 838 m
Formation: Shale Top: 838 m Bottom: 2020 m
Formation: Sandstone Top: 2020 m Bottom: 3271 m
Formation: Limestone Top: 3271 m Bottom: 3690 m

6. DRILLING PARAMETERS BY FORMATION

Representative values summarised from the depth-indexed parameter record for this well. Mud weight is reported in ppg, rate of penetration in m/hr, rotary speed in rpm and weight on bit in klbf.

Formation	Interval (m)	Mud Wt (ppg)	MW range	ROP (m/hr)	RPM	WOB (klbf)
Claystone	0–838	9.60	9.0–10.1	31.7	141	8.3
Shale	838–2,020	11.13	10.3–12.0	15.9	125	17.9
Sandstone	2,020–3,271	12.63	11.7–13.6	15.2	111	25.9
Limestone	3,271–3,690	14.20	13.9–14.5	7.2	90	35.9

6.1 Mud Programme Summary

Water-based inhibited mud was used throughout. Mud weight was raised progressively with depth from approximately 9.6 ppg in the upper section to 14.2 ppg at total depth. Mud weight was trimmed to the minimum safe value across the recorded loss intervals and lost-circulation material was maintained in the active system.

6.2 Bit / Hydraulics Note

Rotary speed was held in the 120 rpm range on average with weight on bit averaging 20.5 klbf. Penetration rate declined with depth from 31.7 m/hr in the upper hole to 7.2 m/hr at total depth, consistent with the area trend.

7. DRILLING EVENTS RECORD

All reportable events for this well, in depth order. Each event is recorded against the formation interval containing the event depth.

Event ID	Depth (m)	Formation	Event Type	Severity	Duration (hr)
E0043	412	Claystone	Wellbore Instability	High	25.4
E0044	524	Claystone	Cementing Issue	High	18.9
E0046	1,180	Shale	Cementing Issue	Medium	10.2
E0045	1,673	Shale	Wellbore Instability	High	23.9
E0047	2,669	Sandstone	Mud Loss	High	11.1
E0049	2,735	Sandstone	Mud Loss	High	16.2
E0048	3,513	Limestone	Wellbore Instability	Critical	36.8
E0050	3,637	Limestone	Casing Issue	Low	5.6

Well ID: W008		Event Type: Wellbore Instability		Depth: 412 m		Formation: Claystone		Severity: High		Duration: 25.4 hours
Well ID: W008		Event Type: Cementing Issue		Depth: 524 m		Formation: Claystone		Severity: High		Duration: 18.9 hours
Well ID: W008		Event Type: Cementing Issue		Depth: 1180 m		Formation: Shale		Severity: Medium		Duration: 10.2 hours
Well ID: W008		Event Type: Wellbore Instability		Depth: 1673 m		Formation: Shale		Severity: High		Duration: 23.9 hours
Well ID: W008		Event Type: Mud Loss		Depth: 2669 m		Formation: Sandstone		Severity: High		Duration: 11.1 hours
Well ID: W008		Event Type: Mud Loss		Depth: 2735 m		Formation: Sandstone		Severity: High		Duration: 16.2 hours
Well ID: W008		Event Type: Wellbore Instability		Depth: 3513 m		Formation: Limestone		Severity: Critical		Duration: 36.8 hours
Well ID: W008		Event Type: Casing Issue		Depth: 3637 m		Formation: Limestone		Severity: Low		Duration: 5.6 hours

7.1 EVENT DETAIL (1–3 OF 8)

EVENT E0043 — WELLBORE INSTABILITY AT 412 m (CLAYSTONE)

Well ID: W008
Event ID: E0043
Event Type: Wellbore Instability
Depth: 412 m
Formation: Claystone
Severity: High
Duration: 25.4 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Sloughing Claystone at 412 m produced excess cuttings volume and hole enlargement; mud properties adjusted. Parameters recorded around the event depth: mud weight 9.62 ppg, ROP 30.6 m/hr, 141 rpm, WOB 8.5 klb.

Response. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.

Outcome. Hole condition stabilised and cuttings volume returned to normal. Interval drilled to section TD without further deterioration.

Historical observation. Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.

EVENT E0044 — CEMENTING ISSUE AT 524 m (CLAYSTONE)

Well ID: W008
Event ID: E0044
Event Type: Cementing Issue
Depth: 524 m
Formation: Claystone
Severity: High
Duration: 18.9 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Poor cement bond opposite Claystone at 524 m; channelling indicated on the bond log and a remedial squeeze planned. Parameters recorded around the event depth: mud weight 9.77 ppg, ROP 31.6 m/hr, 142 rpm, WOB 9.1 klb.

Response. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.

Outcome. Acceptable bond and annular isolation confirmed after the remedial job. Slurry design and displacement rate revised for the next stage.

Historical observation. Annular coverage is sensitive to losses and centralisation across the shoe interval.

EVENT E0046 — CEMENTING ISSUE AT 1,180 m (SHALE)

Well ID: W008
Event ID: E0046
Event Type: Cementing Issue
Depth: 1180 m
Formation: Shale
Severity: Medium
Duration: 10.2 hours
Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Poor cement bond opposite Shale at 1180 m; channelling indicated on the bond log and a remedial squeeze planned. Parameters recorded around the event depth: mud weight 10.79 ppg, ROP 16.5 m/hr, 129 rpm, WOB 16.3 klb.

Response. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.

Outcome. Acceptable bond and annular isolation confirmed after the remedial job. Slurry design and displacement rate revised for the next stage.

Historical observation. Annular coverage is sensitive to losses and centralisation across the shoe interval.

7.2 EVENT DETAIL (4–6 OF 8)

EVENT E0045 — WELLBORE INSTABILITY AT 1,673 m (SHALE)

Well ID: W008
Event ID: E0045
Event Type: Wellbore Instability
Depth: 1673 m
Formation: Shale
Severity: High
Duration: 23.9 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Borehole collapse tendency in Shale at 1673 m; repeated back-reaming needed to maintain hole condition. Parameters recorded around the event depth: mud weight 11.50 ppg, ROP 15.3 m/hr, 125 rpm, WOB 20.6 klbf.

Response. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.

Outcome. Hole condition stabilised and cuttings volume returned to normal. Interval drilled to section TD without further deterioration.

Historical observation. Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.

EVENT E0047 — MUD LOSS AT 2,669 m (SANDSTONE)

Well ID: W008
Event ID: E0047
Event Type: Mud Loss
Depth: 2669 m
Formation: Sandstone
Severity: High
Duration: 11.1 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Seepage-to-partial mud losses observed while drilling Sandstone at 2669 m; mud weight trimmed and losses monitored. Parameters recorded around the event depth: mud weight 12.54 ppg, ROP 16.7 m/hr, 116 rpm, WOB 24.7 klbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

EVENT E0049 — MUD LOSS AT 2,735 m (SANDSTONE)

Well ID: W008
Event ID: E0049
Event Type: Mud Loss
Depth: 2735 m
Formation: Sandstone
Severity: High
Duration: 16.2 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Partial loss of drilling fluid recorded at 2735 m in the Sandstone interval; returns reduced and LCM pumped to regain circulation. Parameters recorded around the event depth: mud weight 12.61 ppg, ROP 16.0 m/hr, 107 rpm, WOB 25.9 klbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

7.3 EVENT DETAIL (7–8 OF 8)

EVENT E0048 — WELLBORE INSTABILITY AT 3,513 m (LIMESTONE)
Well ID: W008 Event ID: E0048 Event Type: Wellbore Instability Depth: 3513 m Formation: Limestone Severity: Critical Duration: 36.8 hours Operational Impact: Major non-productive time; section objectives affected and specialist intervention required. Description. Hole instability and cavings from the Limestone interval at 3513 m; tight hole on connections and reaming required. Parameters recorded around the event depth: mud weight 14.29 ppg, ROP 6.2 m/hr, 89 rpm, WOB 36.5 klbf. Response. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval. Outcome. Hole condition stabilised and cuttings volume returned to normal. Interval drilled to section TD without further deterioration. Historical observation. Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.
EVENT E0050 — CASING ISSUE AT 3,637 m (LIMESTONE)
Well ID: W008 Event ID: E0050 Event Type: Casing Issue Depth: 3637 m Formation: Limestone Severity: Low Duration: 5.6 hours Operational Impact: Minor non-productive time; operations resumed after routine parameter adjustment. Description. Restricted casing passage through plastic Limestone at 3637 m; hole conditioned and casing rerun. Parameters recorded around the event depth: mud weight 14.46 ppg, ROP 6.8 m/hr, 82 rpm, WOB 36.6 klbf. Response. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing. Outcome. Casing landed at the planned shoe depth and tested satisfactorily. Running speed limited through the affected interval on subsequent strings. Historical observation. Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.

8. CASING AND CEMENTING RECORD

Casing	Grade / Weight	Shoe Depth (m)	Hole Size	Cement Programme
13-3/8"	K-55, 68 lb/ft	840	17-1/2"	Lead 12.6 ppg / Tail 15.8 ppg, cement to surface
9-5/8"	N-80, 47 lb/ft	2,020	12-1/4"	Lead 12.9 ppg / Tail 15.8 ppg, TOC 720 m
7"	N-80, 29 lb/ft	3,450	8-1/2"	Tail 15.8 ppg, TOC 1,870 m
5" liner	N-80, 18 lb/ft	3,690	6-1/8"	Tail 15.8 ppg, liner lap tested

8.1 Cementing Observations

The following casing/cementing events were recorded and are cross-referenced to Section 7:

- E0044** — Cementing Issue at 524 m in Claystone (High, 18.9 hr). Poor cement bond opposite Claystone at 524 m; channelling indicated on the bond log and a remedial squeeze planned.
- E0046** — Cementing Issue at 1,180 m in Shale (Medium, 10.2 hr). Poor cement bond opposite Shale at 1180 m; channelling indicated on the bond log and a remedial squeeze planned.
- E0050** — Casing Issue at 3,637 m in Limestone (Low, 5.6 hr). Restricted casing passage through plastic Limestone at 3637 m; hole conditioned and casing rerun.

Remedial work was carried out as described in Section 7 and annular isolation was confirmed before drilling ahead.

8.2 Pressure Testing

Each string was pressure tested after landing and formation integrity tests were carried out below the 13-3/8" and 9-5/8" shoes. Recorded results were satisfactory except where noted in Section 7.

9. FINAL SUMMARY

Final Depth	3,690 m
Primary Formation	Sandstone
Major Events	6 of 8 events rated High or Critical
Total NPT	148.1 hours
Completion Status	Drilled
Offset Risk Classification	HIGH — repeated significant events

9.1 Drilling Challenges

- Wellbore Instability** at 412 m, 1,673 m, 3,513 m (Claystone, Shale, Limestone) — 86.1 hr cumulative.
- Cementing Issue** at 524 m, 1,180 m (Claystone, Shale) — 29.1 hr cumulative.
- Mud Loss** at 2,669 m, 2,735 m (Sandstone) — 27.3 hr cumulative.
- Casing Issue** at 3,637 m (Limestone) — 5.6 hr cumulative.

9.2 Key Historical Observations

- Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.
- Annular coverage is sensitive to losses and centralisation across the shoe interval.
- Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.
- Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.
- Formation tops in this well correlate closely with the neighbouring wells listed in Section 4, supporting depth-based offset comparison across the cluster.

9.3 Lessons Learned

- Raise inhibition before entering the Claystone/Shale/Limestone section and limit open-hole exposure time; treat rising cavings volume as an early warning.
- Review slurry design, centralisation and displacement rate for the stage covering 524 m; run a bond log before proceeding.
- Approach the Sandstone interval (2,669–2,735 m) with LCM pre-mixed and the minimum safe mud weight; monitor returns continuously through this window.
- Allow a dedicated conditioning trip before running casing across 3,637–3,637 m and limit running speed through the tight interval.

10. DOCUMENT CONTROL

Document Number	WCR-W008-R0
Revision	R0
Issue Date	13-Apr-2018
Prepared By	Drilling Engineering (prototype dataset generator)
Classification	Prototype Training Record — synthetic demonstration data
Source Datasets	wells.csv, well_formations.csv, drilling_events.csv, drilling_parameters.csv